

autonomy_light interface

autonomy_light 는 하나의 ROS 2 node다. Super-LIO의 global-frame point cloud와 LiDAR odometry를 소비하고, robot-centric elevation map을 발행한다.

```
| Direction | Topic | Type | Frame | |---|---|---|---| | input | /lio/odom | nav_msgs/msg/Odometry | odom or map → LiDAR | | input | /lio/cloud_world | sensor_msgs/msg/PointCloud2 | odom or map | | output | /autonomy_light/odom | nav_msgs/msg/Odometry | global → base_link | | output | /autonomy_light/path | nav_msgs/msg/Path | global | | output | /autonomy_light/height_map_data | autonomy_light/msg/HeightMap | base_link_gravity | | output | /autonomy_light/height_map_quality | autonomy_light/msg/HeightMapQuality | base_link_gravity | | output | /autonomy_light/height_map | sensor_msgs/msg/PointCloud2 | base_link_gravity | | output | /autonomy_light/live_map | sensor_msgs/msg/PointCloud2 | global |
```

autonomy_light publishes odom|map → base_link → base_link_gravity, plus static base_link → lidar_link. A saved map selects Super-LIO relocation_node and therefore uses map; normal operation uses odom. No world frame exists. Mapping uses Super-LIO keyframes, Scan Context candidate search, GICP verification and an iSAM2 pose graph; its corrected map → odom → imu TF remains continuously broadcast. Saved-map localization keeps local LIO in odom and refreshes the NDT → ICP map → odom correction at the configured low rate. /lio/odom and /lio/cloud_world remain in the selected global frame; this package does not duplicate the SLAM backend.

HeightMap.data is reference_height - terrain_height, clamped by height_map_distance. A quality mask marks unobserved cells explicitly: valid=0 means data contains only the configured unknown placeholder.

Cyclone DDS transport

Set height_map_output.transport to cyclone_dds for a direct DDS output, or both to retain the ROS 2 outputs during migration. The installed IDL is resources/idl/HeightMap.idl and defines exactly:

```
module core_dds { struct HeightMap { sequence<float> data; }; };
```

The configurable default is domain 1, topic height_map, type core_dds::HeightMap, best-effort keep-last history depth 1. Its sole field has the same row-major cell order, length and unknown convention as ROS HeightMap.data; it intentionally carries no header, geometry or quality channels. Consumers must share the grid configuration and read the ROS quality stream when that metadata is required.